

Metagenomic Investigations on Antibiotic Resistance and Microbial Virulence in Oil-Polluted Soils from China

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Research Article

Keywords: Engine oil pollution, resistosome, virulome, oil polluted soil, microbial community

Posted Date: August 7th, 2023

DOI: <https://doi.org/10.21203/rs.3.rs-3176132/v1>

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Version of Record: A version of this preprint was published at Environmental Science and Pollution Research on October 4th, 2023. See the published version at <https://doi.org/10.1007/s11356-023-30137-z>.

Abstract

Engine oil spills have been associated with a wide range of human health problems. However, little is known about the effects of petroleum hydrocarbon pollution on soil microbial communities. In this study, three samples were collected from oil-polluted soils (OPS) and one control soil (CS), from Taolin town, China, near the old engine's scrapes were used. The aims of this study were to conduct metagenomic sequencing and subsequently perform resistome and Virulome analysis. We also aimed to validate antimicrobial resistance and virulence genes and antibacterial sensitivity profiles among the isolates from oil polluted soils. The OPS microbial community was dominated by bacterial species compared to the control samples which were dominated by metazoans and other organisms. Secondly, the resistosome and virulome analysis showed that ARGs and virulence factors were higher among OPSs microbial communities. Antibiotic susceptibility assay and qPCR analysis for ARGs and virulence factors showed that the oil polluted soil samples had remarkably enhanced expression of these ARGs and virulence genes. Our study suggests that oil pollution contributes to shifting microbial communities to more resilient types that could survive the toxicity of oil pollution and subsequently became more resilient in terms of higher resistance and virulence potential.

Introduction

Oil contamination of soils is a significant environmental problem with far-reaching consequences (Shi et al. 2023). Oil spills and leaks in its immediate surrounding causes ecological perturbation which is detrimental to both human and animal health (Salam et al. 2021). Further oil spills can also influence the development of antimicrobial resistance (AMR) and microbial virulence by exerting selective pressure (Salam et al. 2021). The phenomenon not only affects the immediate surroundings but also extends to other ecosystems such as water bodies and agricultural farms (Shi et al. 2023). As a consequence, it can potentially increase the incidence and severity of antimicrobial resistant infectious diseases (Lin et al. 2023, Shi et al. 2023).

Antimicrobial resistance, microbial virulence and oil contamination are a triple threat and a growing modern problem to the One Health. Antimicrobial resistance mechanisms to various environmental stress is responsible for resilience in strains that enable them to survive the toxic environment (Chunyan et al. 2022, Zhu et al. 2022). Engine oil contains various heavy metals such as Pb, Cu, Cr, Ni and Fe concentrations and polycyclic aromatic hydrocarbons (PAHs) (Kashif et al. 2018, Szyszlak-Bargłowicz et al. 2021). The presence of heavy metals and PAHs exerts selective pressure on microbiota in the polluted environment that leads to the development of antibiotic-resistant bacteria by mechanisms such as cross resistance (e.g., efflux pumps that enables resistance to antibiotics and heavy metals, biocides, etc) and co-resistance (co-location of antibiotic and heavy metal resistance genes on a single mobile genetic element) (Gothwal &Shashidhar 2015) (Suhadolnik et al. 2022).

Environment pollution has been strongly linked to the enhancement of antimicrobial resistance and virulence in microbes (Quillaguaman et al. 2021, Zhang et al. 2022).

Oil spills and leaks in soil can also enrich opportunistic pathogens (Lin et al. 2023). However, studies investigating the roles of oil pollution on preponderance of antimicrobial resistance and microbial virulence factors are sparse. Here, we investigated the perturbation of virulence and resistance complement of soil microbial communities in engine oil-contaminated soil samples. The aims of this study were to conduct metagenomic sequencing and subsequently perform resistome and virulome analysis. We also aimed to validate antimicrobial resistance genes (ARGs) and virulence factors using molecular techniques. Studies such as these are crucial for understanding the ecological impact, assessing risks to human health, guiding remediation efforts, and monitoring the spread of resistance and virulence in the environment.

Materials and Methods

Site description and sample collection

Three samples were collected from oil-polluted soils (OPS), and one control soil (CS) was obtained from an old engine scrap plant in Taolin Town, Donghai County, Lianyungang City, Jiangsu Province, China. OPS sample one was obtained from a dry soil polluted with engine oil, located between latitude 34°31'2"N and 118°29'45"E, at an elevation of 60 meters above sea level. OPS sample two was collected from soil near the streaming water polluted with oil, located between latitude 34°31'21"N and 118°29'29"E, at an elevation of 56 meters above sea level.

OPS sample three was obtained from the soil of a stagnating pool of engine oil, located between latitude 34°31'23"N and 118°28'58"E, at an elevation of 62 meters above sea level. The CS sample, used as a control, was collected at a depth ranging from 0–6 cm, between latitude 34°31'20"N and 118°29'29"E, at an elevation of 60 meters above sea level. Each site was sampled in triplicate using sterile, DNAs- and RNAs-free tubes (Corning®, USA). The samples were immediately transferred to the lab for total DNA extraction and bacterial culture, while the remaining samples were stored at -80°C until further processing.

Soil hydrocarbons test

To study the petroleum hydrocarbon contents in OPS, about 10g of sample, was mixed with an appropriate amount of anhydrous sodium sulfate to dehydrate the sample. The procedures were modified from the previous study (Su et al. 2023). Briefly, the samples were transferred to the filter cartridge and 100mL of n-hexane-acetone mixed reagent (1:1) was added and extracted for 16-18h. After cooling, the extract was concentrated and purified and finally concentrated to 1ml for testing by gas chromatography GC-ECD/FID TRACE1300 (Thermo Fisher Scientific, USA). The experimental conditions were as follows: The theoretical concentration value of QC is: 1550 µg/mL; Sample spiked quantity: 1550 µg; Blank standard: 1550ug; Standard solution No.: LH220825-01. Sample inlet temperature: 280 °C, column oven temperature: 50 °C for 2 min, rise to 230 °C at 40 °C/min, and then rise to 300 °C at 20 °C/min for 5 min, chromatographic column model: select mineral oil, detector temperature: 300 °C.

Total DNA extraction, library construction and Shotgun metagenomic sequencing

DNA from the soil samples (OPS1, OPS2, OPS3, and CS) were extracted and purified using the E.Z.N.A. Soil DNA Kit D5625 (Omega Bio-Tek, Inc., USA) as per the manufacturer's instructions. The protocol was modified according to a previous report (Bao et al. 2017). Briefly, 5 grams of soil was placed in a 50 ml sterile polypropylene tube and resuspended in 10 ml Buffer SLX Plus PD09 with 5 g of glass beads (0.1 mm). The mixture was then agitated in a mini-bead beater, thrice for 60 seconds each time. After that 1 ml Buffer, DS was added, the samples were vortexed and incubated at 70°C for 10 min followed by 90°C incubation for 5 min. The quality check of extracted DNA was performed using 1% agarose gel electrophoresis and quantified using the Nanodrop 2000 (Thermo Fisher Scientific, USA). The three OPS and CS were sequenced and analyzed Mojobio company (Majorbio, Shanghai, China) for shotgun metagenomic sequencing. Briefly, 40 ng of DNA were used to prepare the libraries with a Nextera DNA kit (Illumina, USA), as per manufacturer's instructions. Afterward, the adapters were added, libraries diluted and pair-end sequenced for 300 cycles using a NovaSeq Reagent/HiSeqX system (Illumina, USA).

Metagenomic analysis

For raw sequence analysis, multiple samples were mixed and sequenced in parallel, and the sequences in each sample were labeled with an Index tag sequence indicating each sample. The data of each sample are distinguished according to the Index sequence, and the extracted data is saved in fastq format. To perform data quality control the low-quality and N-containing reads were trimmed. Prodigalv2.6.3 (Hyatt et al. 2010) was used for gene prediction. Genes whose length is greater than or equal to 100bp and translated into amino acid sequences.

The predicted gene sequences of all samples were analyzed by CD-HIT software (Fu et al. 2012) for clustering, each class taking the longest gene as a representative sequence to construct a non-redundant gene set. To calculate gene abundance, we used SOAP aligner software to compare the high-quality reads of each sample with the non-redundant gene (NR) set and count the abundance information of the genes in each sample. NR species annotation from NR database using Amino acid sequence database of non-redundant proteins, including SwissProt, PIR (Protein Information Resource), PRF (Protein Research Foundation), PDB (Protein Data Bank) protein database non-redundant data and protein data translated from GenBank and RefSeq CDS data. To compare the non-redundant gene set with the NR database (comparison type: BLASTP), DIAMOND software (Buchfink et al. 2015) was used and passed the corresponding classification of the NR database. In the taxonomic information database, we compared the gene abundance of species in each sample at each taxonomic level of domain, kingdom, phylum, class, order, family, genus, and species in order to construct the abundance profile at the corresponding taxonomic level. Comprehensive Antibiotic Resistance Database (CARD) was used to analyze the antibiotic resistance genes (Alcock et al. 2023). The virulence factors among the microbial communities isolates from oil samples were analyzed using the Virulence Factor Database (VFDB) (Chen et al. 2005)

and for comparison with data from VFDB, we used DIAMOND software to determine the virulence factors genes (VFGs).

Isolation of oil-degrading microbes and characterization

To isolate oil-degrading microbes we took 5g of soil samples and added it to 5 ml Bushnell-Haas (BH) medium supplemented with 1% engine oil and incubated in a shaker incubator at 37°C with 180rpm/min for 7 days.

The turbid bacterial culture (one ml) was transferred to a fresh BH agar medium supplemented with 1% filtered oil. Single colonies were selected, and DNA was extracted from 10 candidate bacterial strains. 16S rRNA sequencing using the primer pairs; 27F and 1492R (Suppl. table 1) was performed at Majorbio, Shanghai, China. The obtained sequences were trimmed, aligned and analyzed using the EzBioCloud16S rRNA tool (Yoon et al. 2017), and NCBI-BLASTn to identify the corresponding microbes.

Quantitative PCR and gene expression analysis.

For the qRT-PCR analysis, we followed the method previously described (Qaria et al. 2018). Briefly, RNA was isolated from *P. aeruginosa*, *A. marplatensis*, *K. pneumoniae* and *Leclercia sp.* By TRIzol (Invitrogen, USA). Subsequently, 3 µg of RNA was converted to cDNA using SuperScript-III (Invitrogen, USA) and random hexamers according to the manufacturer's instructions. For qRT-PCR, 40 ng of the first transcribed DNA strand was amplified by using SYBR Fast qPCR Mix (TaKaRa, Japan) with primers targeting *iroC*, *ybtQ*, *wzt*, *gmd*, *clbM* genes and 16S rRNA as an internal control. The primer sequences are listed in (Suppl. table 1).

Antibiotic susceptibility testing for microbes isolated from oil polluted soil.

Antimicrobial sensitivity of 4 strains isolated from OPSs and control sample against 9 different antibiotics as published previously (Ranjan et al. 2016). Briefly, was performed on Mueller-Hinton agar (MHA) plates by the M.I.C.E.™ (Oxoid, USA), according to the manufacturer's protocols. *P. aeruginosa*, *A. marplatensis*, *K. pneumoniae* and *Leclercia sp.* strains grown on Mueller-Hinton agar medium were collected and washed with PBS.

About 50 µl of 10^8 of *P. aeruginosa*, *A. marplatensis*, *K. pneumoniae* and *Leclercia sp.* cells suspension was spread on Mueller-Hinton agar medium M.I.C.E.™ corresponding to gentamicin, kanamycin, cefoperazone, polymyxin B, ciprofloxacin, sulfamethoxazole, rifampicin, cefradine and ampicillin were placed on inoculum spread plates and incubated for 16–24 hours. The susceptibility was interpreted by breakpoints defined by the Clinical and Laboratory Standards Institute (CLSI 2022).

Statistical analysis

The statistical analyses were performed using one-way ANOVA followed by Tukey's multiple-comparison tests wherein ns denotes to non-significant * denotes $P < 0.05$ ** denotes $P < 0.01$, *** denotes $P \leq 0.001$ and ****denotes $P \leq 0.0001$. The data were presented as mean $\pm$ the standard error (of mean) from three

independent experiments. Graphs were plotted either by OriginPro software (OriginLab, USA) or GraphPad Prism 8.0.2 software (Dotmatics, USA)

Results

Petroleum hydrocarbons in contaminated soils

The levels of petroleum hydrocarbons among the OPSs were analyzed and compared to the levels of hydrocarbons in the control soil. The results showed that the OPS-1 isolated from the soil near the scrap plant contained the highest petroleum hydrocarbons of 846 mg/kg followed by OPS-2 (261 mg/kg) and OPS-3 (223 mg/kg). The control sample demonstrated a relatively lower concentration of petroleum hydrocarbons which was 156 mg/kg (Table 1).

Table 1
Levels of petroleum hydrocarbons determined in the soil samples.

Soil test	CS	OPS-1	OPS-2	POS-3
Petroleum hydrocarbons (C10-C40) mg/kg	156	846	261	223

Microbial diversity of oil polluted soils

To determine the microbial communities (taxa) among OPSs, metagenomic sequences were analyzed in each sample. OPS-1 was dominated by bacterial species and interestingly contained the largest archaea communities among all samples. OPS-1 and OPS-2 were dominated by bacterial species compared to the CS which was dominated by metazoan species (Fig. 1a). The phylum analysis for microbial community isolated from OPSs showed the dominance of Actinobacteria, Acidobacteria and less diverse on other organism phyla, while CS significantly dominated with Proteobacteria and rich in other organisms' phyla such as Arthropods (Fig. 1b). Interestingly, data showed that the Actinobacteria community was significantly less in CS and high percentage of proteobacteria compared to the ones in OPSs. Overall, the OPSs organism communities were dominated by bacterial species compared to the CS which were dominated by metazoan and other organisms.

Resistome of oil-polluted soil microbiome

To check the antibiotic resistance among the OPSs microbial communities the antibiotic resistosome was analyzed using the CARD database. The results showed that the OPS-3 microbial community was remarkably highly resistant to several classes of antibiotics as it harbored genes conferring resistance to multiple antibiotics and in abundance, this sample was obtained from the stagnant pool of engine oil.

While the other two samples, OPS-1 and OPS-2 also revealed diverse resistance genotypes conferring resistance to multiple antibiotic classes. Comparatively, antibiotic resistance of the CS microbial community was at the baseline, but the preponderance was least compared to the OPSs. Overall, the OPS-3 microbial community were enriched with AMR genotypes belonging to multiple antibiotic classes

compared to the other samples (OPS-1, OPS-2 and CS) (Fig. 2a&b). The analysis of antibiotic resistance mechanisms using the CARD revealed the antibiotic efflux mechanisms to be the most common strategy among the OPS microbial community (Fig. 2c&d). As with the previous observation of an abundance of AMR genotypes, the OPS-3 microbial community demonstrated high prevalence of efflux mechanisms.

Virulome of oil-polluted soil microbiome

The virulome of metagenomic samples was determined using the VFDB and DIAMOND tools to predict the VFGs abundance and functional analysis. The analysis showed that the virulome of the OPS-3 microbiome that was isolated from the oil stagnant pool showed abundant virulence factors belonging to diverse mechanisms, while the soil microbiome (OPS-2) of nearby running polluted water harbored the least virulence determinants among the OPSs (Fig. 3a). The virulome of the control microbiome showed a remarkably less virulence prevalence in contrast to the OPSs microbiome. Iron uptake system, toxins, secretion system and adherence strategies were among the most common virulence mechanisms detected among the OPS microbiomes (Fig. 3b). From the above observation, the soil samples exposed to petroleum hydrocarbon pollution showed higher prevalence of virulence genotypes compared to less polluted soil and control.

Bacterial strains isolation and characterization

We purified a single colony from each soil sample to perform in vitro analysis of the antibiotic resistance and virulence of antibiotics. The purified colonies were subjected to 16S rRNA sequence analysis. Based on the 16S rRNA sequence analysis using NCBI-BLASTn results showed that the 16S rRNA sequence matched with *Leclercia sp*, *P. aeruginosa*, and *K. pneumoniae* from OPS-1, OPS-2 and OPS-3 respectively. The strain isolated from CS showed high similarities with *A. marplatensis*.

Oil pollution renders bacteria resistant to antibiotics

To investigate the effect of oil pollution on antibiotic resistance of the microbial community. We determined MICs of several antibiotics on strains isolated from OPSs and CS samples.

Based on the results of 16S rRNA sequence analysis, one bacterial strain from each sample was selected for antimicrobial susceptibility testing (AST). *A. marplatensis*, isolated from the control sample was used as a reference bacterial strain, *Leclercia sp*, *P. aeruginosa*, and *K. pneumoniae* from OPS-1, OPS-2 and OPS-3, respectively were used for AST. *P. aeruginosa*, *A. marplatensis*, *K. pneumoniae* and *Leclercia sp*. were found to be resistant to several of the tested antibiotics (Table 2). Moreover, to further validate the expression of antibiotic resistance among the bacterial strains isolated from the OPSs, the mRNA expression levels of antibiotic-resistance determinants, *tetA* and *bla_{CMY}* were analyzed (Fig. 4a&b). *tetA* mRNA expression level was significantly upregulated in all strains (*P. aeruginosa*, *K. pneumoniae* and *Leclercia sp*) isolated from the OPSs compared to *A. marplatensis* which was isolated from the control sample. Increased folds of *bla_{CMY}* expression was also seen in bacteria isolated from the OPSs compared to bacteria from control soil.

Overall, the oil contaminated soils exhibits microbial resistance to the antibiotics, which in turn could lead to the dissemination of AMR in the surrounding ecosystem.

Table 2
MICs of *P. aeruginosa*, *K. pneumoniae* and *Leclercia sp.* strains isolated from OPS and *A. marplatensis* isolated from CS sample.

Antibiotic mcg/ml	<i>A. marplatensis</i>	<i>P. aeruginosa</i>	<i>K. pneumoniae</i>	<i>Leclercia sp.</i>
Gentamicin	≤ 16 (S)	≥ 100 (R)	≥ 100 (R)	≥ 50 (R)
Kanamycin	≤ 16 (S)	≥ 30 (R)	≥ 30 (R)	≤ 10 ± 5 (S)
Cefoperazone	≤ 16 (S)	≤ 8	≥ 30(R)	≤ 8
Polymyxin B	≥ 256 (R)	≥ 256 (R)	≥ 256 (R)	≥ 256 (R)
Ciprofloxacin	≤ 0.064 (S)	≤ 0.064 (S)	≤ 1.50 (R)	≤ 1.5 (R)
Sulfamethoxazole	≤ 2 (S)	≤ 2 (S)	≤ 2 (S)	≤ 2 (S)
Rifampicin	≥ 5 (R)	≥ 5 (R)	≥ 5 (R)	≥ 5 (R)
Cefradine	≤ 8 (S)	≥ 30 (R)	≥ 30 (R)	≥ 30 (R)
Ampicillin	≥ 10 (R)	≥ 20 (R)	≥ 20 (R)	≥ 20 (R)

Oil pollution renders upregulation of virulence mechanisms.

To demonstrate the virulence abilities among the isolated strains from the petroleum hydrocarbons polluted soils we analyzed the gene expression of some virulence-related genes. The *E. coli* DH5α was used as a control strain, *P. aeruginosa*, *K. pneumoniae* and *Leclercia sp.* strains from oil polluted samples and *A. marplatensis* from control soil sample was analyzed. The mRNA expression levels of genes, *iroC*, *ybtQ*, *wzt*, *gmd*, and *clbM* were analyzed using qPCR analysis (Fig. 5a-e).

Interestingly, *wzt*, *gmd*, and *clbM* mRNA levels were upregulated in *K. pneumonia* that was isolated from OPS-3. The *iroC*, *wzt*, *gmd*, and *clbM* mRNA expression levels were relatively high in *Leclercia sp.* isolated from OPS-1 compared to the control strain. The *iroC* mRNA expression levels were remarkably higher in *P. aeruginosa* isolated from OPS-2 compared to other strains. Furthermore, in *A. marplatensis* isolated from the CS, the *ybtQ* mRNA expression level was remarkably higher compared to other strains. From the above, the bacterial strains isolated from the OPS showed higher levels of virulence genes compared to the bacteria isolated from CS sample.

Discussion

In this study, we systematically analyzed the possible effects of engine oil spills on the soil microbial profile and their resistome and virulome analysis using comprehensive metagenomic analysis. We also shed light on the antimicrobial sensitivity profiles among the isolates from OPSs and further studied the

ARGs expression and virulence genes expression. Engine oil contains various heavy metals such as Pb, Cu, Cr, Ni and Fe and their concentrations vary between fresh and used engine oils (Kashif et al. 2018, Szyszlak-Bargłowicz et al. 2021).

Mantel test reported that there is a remarkable correlation between heavy metal tolerance and antibiotic resistance (Máthé et al. 2012). Another study showed that bacterial strains isolated from the soil contaminated with heavy metals like As, Co, Cu, Hg, Ni, and Pb exhibited high expression of one or more antibiotic resistance genes (ARGs), such as *bla*_{TEM}, *bla*_{CTX-M}, and *tetA-E* (Knapp et al. 2017). In line with these reports, our results showed that the strains isolated from engine oil-contaminated soils had increased expression of *tetA* and *bla*_{CYM} genes. These observations indicate that the presence of heavy metal oil in the used engine oil might be responsible for the expression of ARGs in the isolated strains.

The concentration of total PAHs detected in the soil samples contaminated with engine oil spells collected from the sites ranged from 0.0184 ± 0.02 to 0.1385 ± 0.2 mg/kg (Obini et al. 2013).

Studies describing the relationship between PAH anthropogenic pollution pressure and ARGs showed that PAHs are closely related to antibiotic resistance and increased mutation rates leading to ARGs (Sazykin et al. 2021). We showed that the strains from OPSs are resistant to several antibiotics. We observed that the soil samples from oil polluted sites had decreased metazoan communities compared to the control sample and were overwhelmingly dominated by bacteria taxa. A previous study also observed similar changes of decrease in the abundance of metazoan taxa in the spill sites compared to pre-spilled sites (Bik et al. 2012). Moreover, through metagenomic analysis, we observed that the ARGs among the OPSs microbial communities were significantly higher, especially those related to multidrug and tetracyclines resistance. The dominant strategies of microbial communities expressed in oil spilled sites were found to be antibiotic efflux and antibiotic target alteration. Similarly, another study demonstrated that the resistance profile of ARGs in the PAHs-polluted soils was characterized by the dominance of efflux pump-encoding genes (Chen et al. 2017). Further, our study demonstrated that strains *P. aeruginosa*, *K. pneumoniae* and *Leclercia sp.* isolated from the OPSs were resistant to multiple antibiotics like polymyxin, ampicillin and gentamycin. Previously, the study showed that *Pseudomonas sp.* isolated from oil-polluted soil was resistant to ampicillin, ceporex and gentamycin compared to the strains from unpolluted sites (Ataikiru et al. 2020). Given this, our results suggest that the engine oil pollution in-deed increases the selective pressure to select antibiotic resistant microbes. This could be to the presence of heavy metals and petroleum hydrocarbons in the soil that enhances the cross selection and co-selection mechanisms that enables them to resist the toxic environment.

Moreover, we observed that the virulence-related genes of the microbial communities from oil polluted soil samples were remarkably higher compared to the microbial communities isolated from the control soil. Most of virulence factors that were highly elevated in OPSs were related to the iron uptake system, secretion system, adherence and regulation. We speculate that oil pollution simultaneously enhances the virulence strategies of microbial communities along with resistance strategies by mechanisms such as co-selection (Hussain et al. 2023). A previous study showed that the bacterial strains isolated from an oil-

contaminated environment exhibited at least one virulence factor along with resistance genes (Cirstea et al. 2021). This is the first study that has analyzed the virulome of oil-polluted microbial communities and examined the expression of virulence-related genes in strains from OPSs.

In conclusion, we conducted a comprehensive analysis of the effects of engine oil spills on soil microbial communities. Our findings revealed significant shifts in the microbial profile, with increased expression of antibiotic resistance genes (ARGs) and virulence-related genes. The presence of heavy metals and petroleum hydrocarbons in the soil likely contributed to the selection of antibiotic-resistant and virulent microbial strains. These observations highlight the importance of considering the environmental impacts of engine oil spills on microbial communities and the potential risks associated with increased resistance and virulence. Further research is needed to identify molecular and genetic mechanisms underlying the development of resistant and virulent microbes in oil-contaminated soils. Such investigations can contribute to the development of effective strategies for mitigating the adverse effects of oil spills on microbial ecosystems and preventing the spread of antibiotic resistant and virulent microbes.

Declarations

Ethical Approval

Not applicable

Consent to Participate

Not applicable

Consent to Publish

All authors agreed with the content and that all gave explicit consent to submit and that they obtained consent from the responsible authorities at the institute/organization where the work has been carried out, before the work is submitted. All authors have read and agreed to the published version of the manuscript.

Author Contributions:

Majjid A Qaria conceptualized the idea, performed the experiments, wrote and prepared the figures and tables **Chunyan Xu** Cured the data, performed the experiment and analyzed the data. **Arif Hussain** improved the manuscript contents wrote, edited, discussed and reviewed the manuscript **Muhammad Zohaib Nawaz** edited, discussed and reviewed the manuscript **Daochen Zhu** supervised, edited and improved the idea of the manuscript.

Funding

This work was supported by the Key Research and Development Program of Jiangsu Province (Grant No. BE2021691), the Natural Science Foundation of Jiangsu Province (Carbon neutralization, BK20220003)

and the Jiangsu Collaborative Innovation Center of Technology and Material of Water Treatment, Suzhou University of Science and Technology, Suzhou 215009, China.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Figures

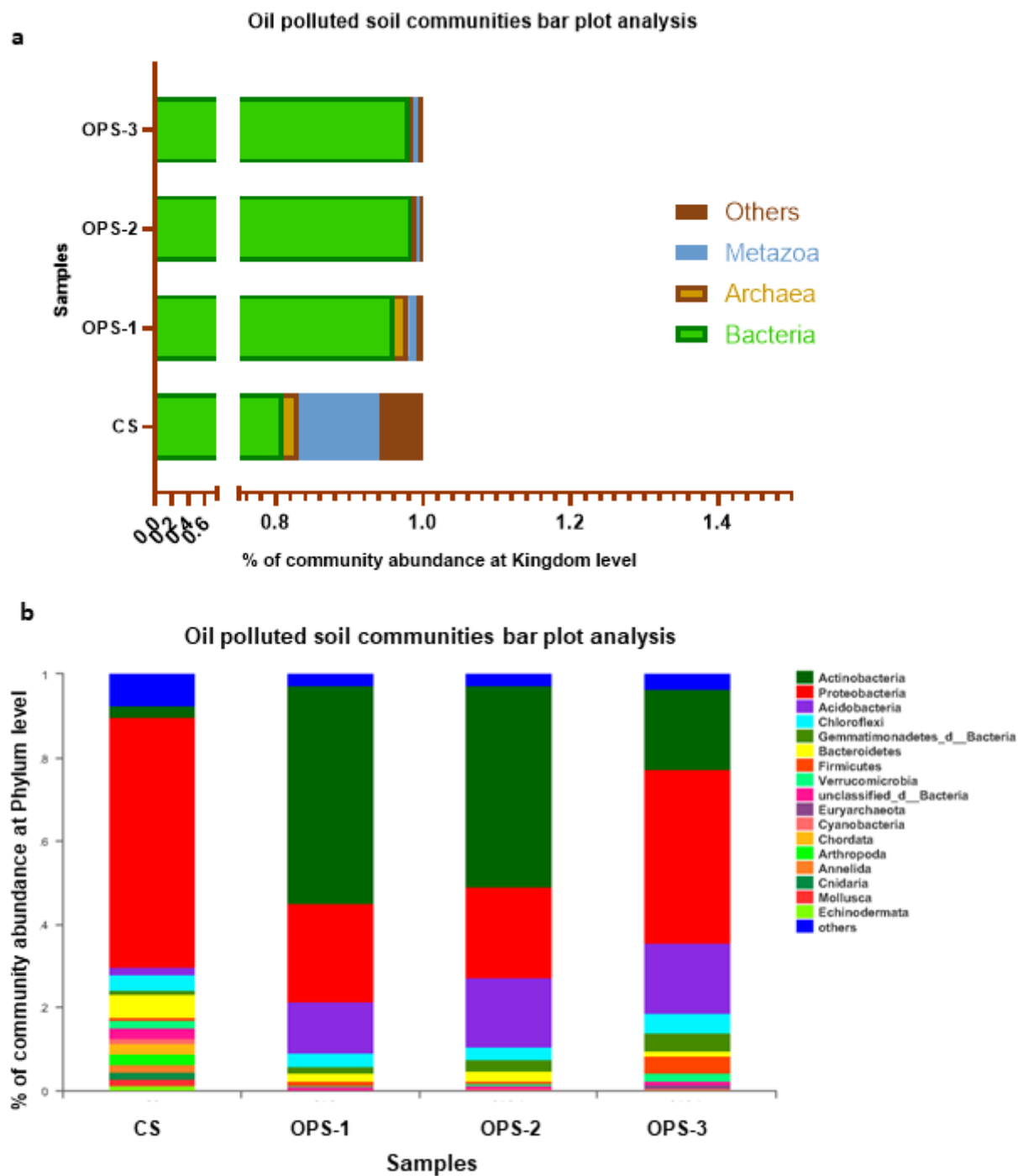


Figure 1

Profiling organism community isolated from three OPS and CS samples using NR database for species and statistical analysis species used to find kingdom abundance (a) and phylum abundance (b).

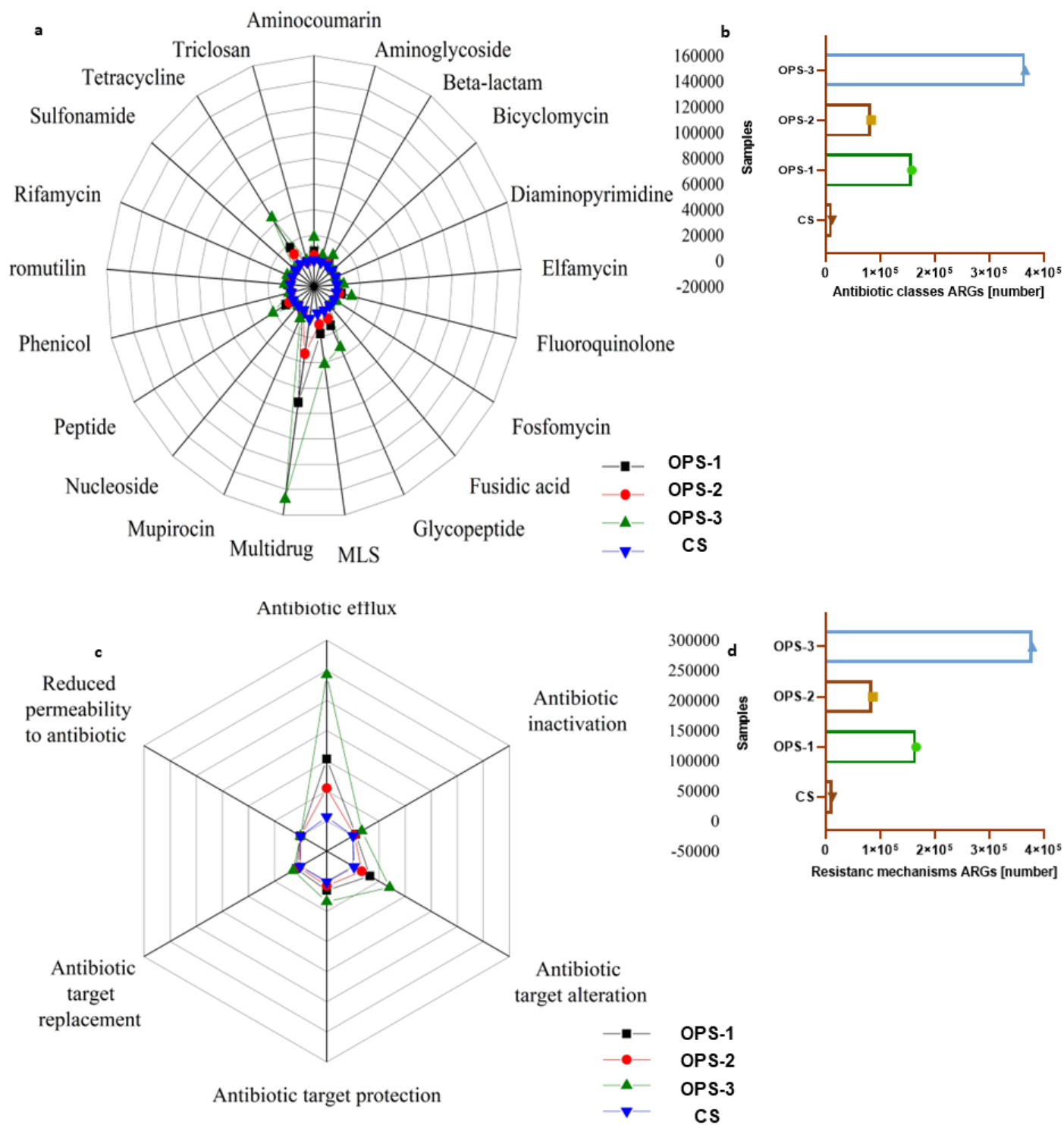


Figure 2

Using CARD tool analysis for the ARGs abundance related to each antibiotic class (a&b) and antibiotic resistance mechanisms (c&d) among the microbes isolated from OPS and CS samples.

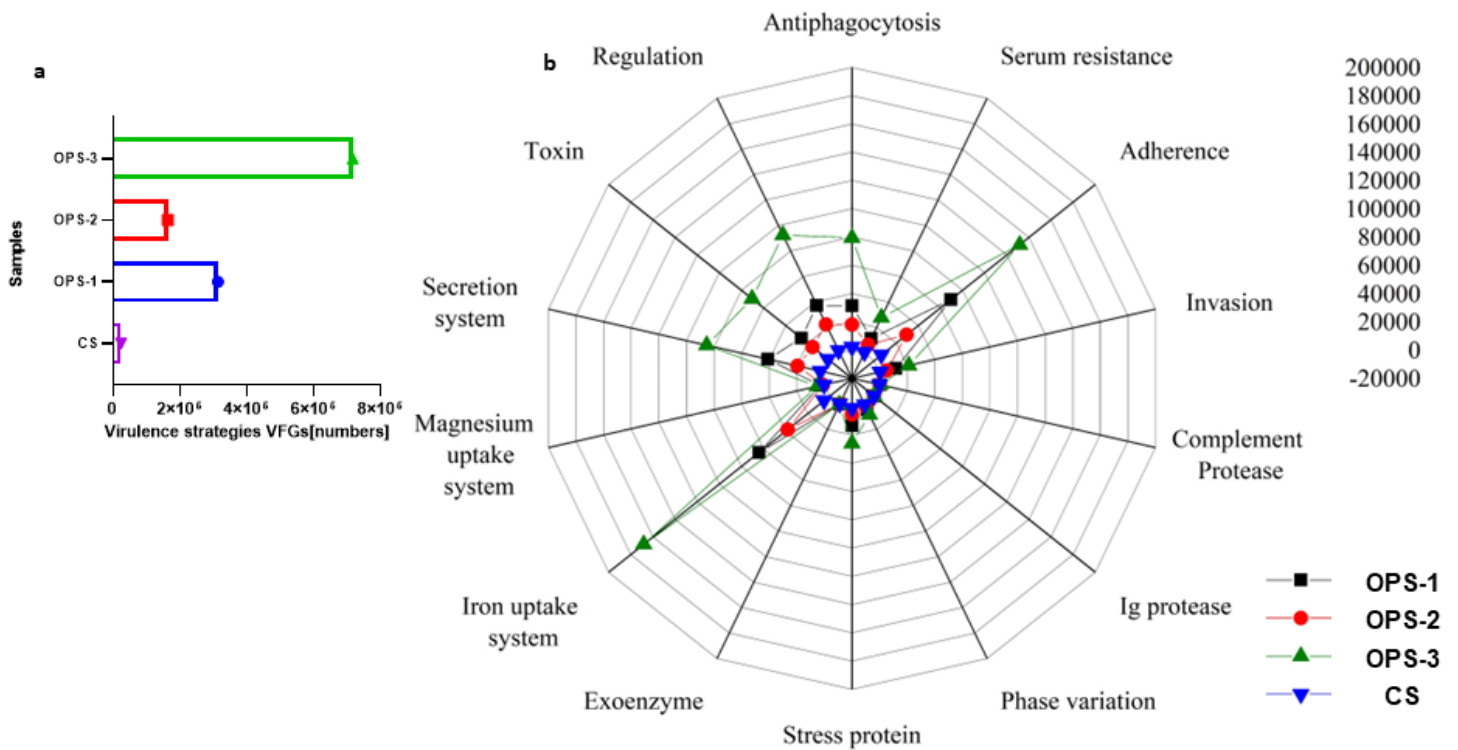


Figure 3

VFGs abundance among the bacterial communities isolated from OPS and CS samples (a), and functional analysis (b) using VFDB and DIAMOND tools.

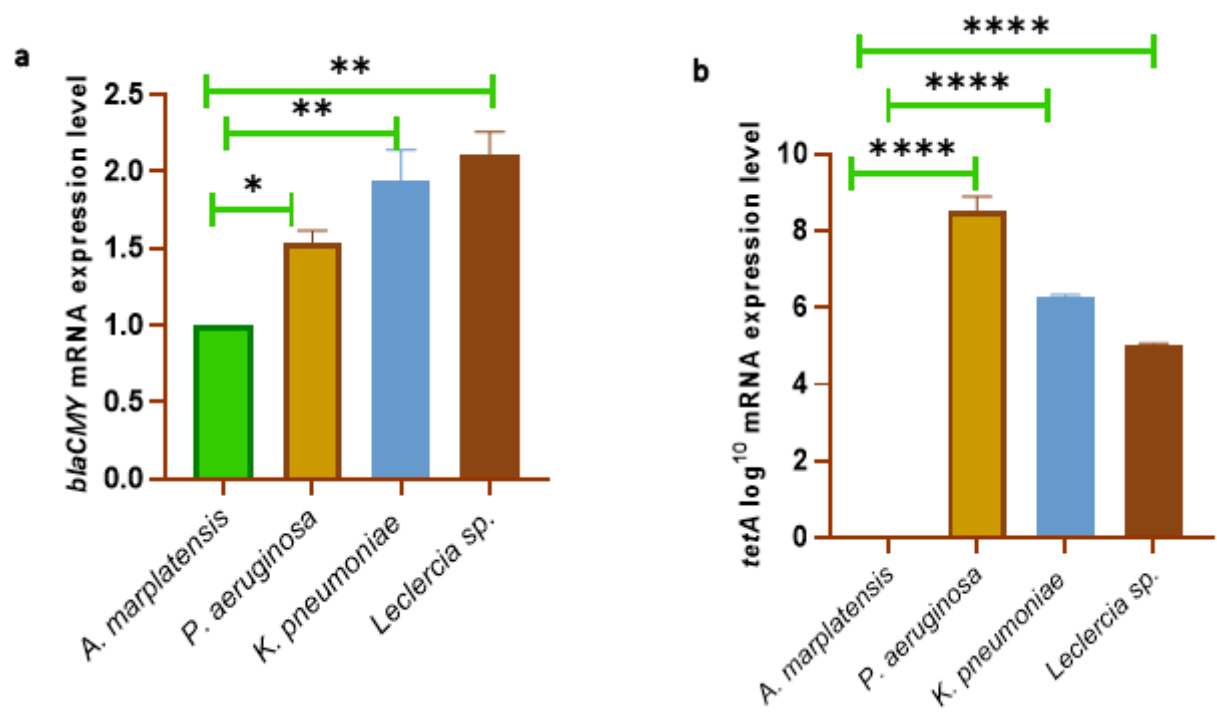


Figure 4

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mRNA level analysis using qPCR for antibiotic resistance genes *bla*CMY (a) and *tet*A(b) among representative strains *P. aeruginosa*, *K. pneumoniae* and *Leclercia* sp. strains isolated from OPS and *A. marplatensis* from CS samples. * denotes $P < 0.05$ ** denotes $P \leq 0.01$ and ****denotes $P \leq 0.0001$.

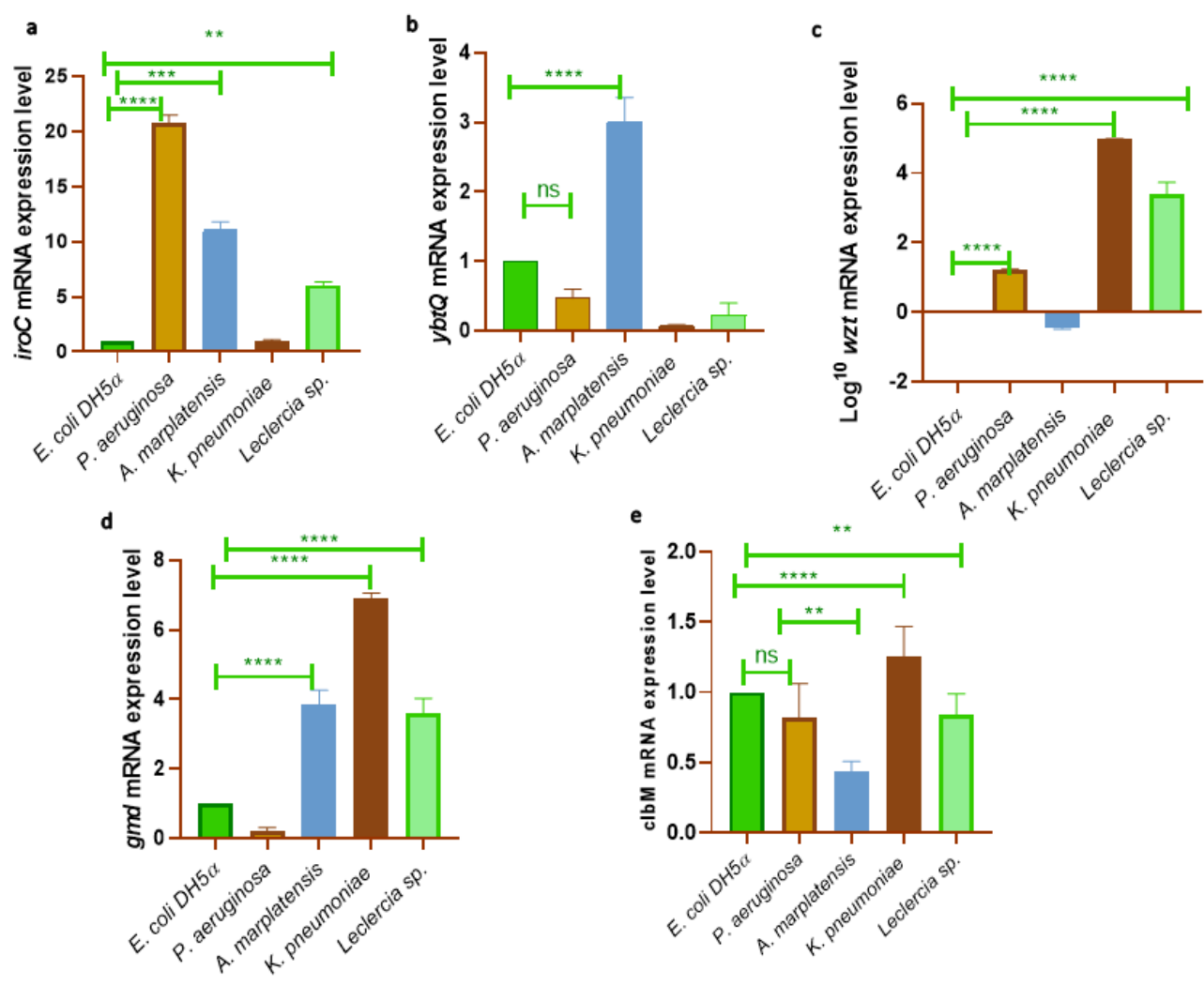


Figure 5

Comparison of mRNA expression level of virulence-related genes including **(a)** *iroC*, **(b)** *ybtQ*, **(c)** *wzt*, **(d)** *gmd*, and **(e)** *clbM* in four representative strain and *E. coli* DH5α mRNA expression levels were analyzed using qPCR. (ns denotes to non-significant * denotes $P < 0.05$ ** denotes $P < 0.01$, *** denotes $P \leq 0.001$ and ****denotes $P \leq 0.0001$).

Supplementary Files

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